Heart Disease Dataset



Albatul Altoum

1. Dataset/Problem Overview

The dataset used in this project is the Heart Disease Dataset from Kaggle, which contains 1,025 records and 14 attributes representing medical and demographic information such as age, cholesterol level, resting blood pressure, and maximum heart rate. The target variable indicates whether a patient has heart disease (1) or not (0). The dataset was downloaded using the opendatasets library, explored with pandas, and prepared for machine learning. This dataset is suitable for binary classification tasks in healthcare to predict the likelihood of heart disease.

2. What I Aim to Solve

The goal of this project is to build a supervised machine learning model capable of predicting the presence of heart disease based on the patient's clinical data. The focus is on achieving a balance between model interpretability and accuracy while creating a reproducible and efficient pipeline.

3. Why It Matters

Heart disease is one of the leading causes of death globally, and early prediction helps healthcare providers identify high-risk patients for interventions by analyzing this dataset, we demonstrate how data-driven methods assist doctors and improve decision-making in preventive medicine.

4. Applied ML Approach

A Logistic Regression model was chosen as the baseline classifier because it is interpretable, efficient, and performs well on binary classification tasks. Before training, data was standardized using the StandardScaler to improve model stability and convergence. Class imbalance was handled by assigning balanced class weights. The model was trained using scikit-learn with a maximum of 1,000 iterations to ensure proper convergence.

5. Model and Methodology

The data preprocessing included cleaning and handling missing cholesterol values by replacing them with the column mean, and encoding categorical variables The dataset was divided into 75% training and 25% testing sets using stratified sampling to maintain class balance Numerical features were standardized using the StandardScaler from scikit-learn A Logistic Regression model from scikit-learn was applied for binary classification due to its interpretability and efficiency The model was trained with balanced class weights and a maximum of 1,000 iterations to ensure convergence Performance was evaluated using accuracy, precision, recall, F1-score, and confusion matrix metrics Finally, the trained model, scaler, and cleaned dataset were saved using the Joblib library for reproducibility.

6. Results and Discussion

The model achieved an accuracy of approximately 83%, which indicates good predictive performance. The confusion matrix showed that the model correctly identified most cases of both healthy and heart disease patients, with few false predictions. The F1-score was balanced, confirming consistent performance across both classes. The boxplot visualization showed that patients with heart disease tend to be older, while the correlation heatmap revealed moderate relationships between features such as chest pain type and heart disease. Histograms of numerical features indicated realistic data distributions. Overall, the logistic regression model provided a strong, interpretable baseline suitable for clinical risk prediction.

Visualization References

The following visualizations were generated to support the report findings:

```
Accuracy: 0.8326848249027238

Classification Report:
              precision    recall  f1-score   support

           0      0.866     0.776     0.819       125
           1      0.807     0.886     0.845       132

    accuracy                          0.833       257
   macro avg      0.836     0.831     0.832       257
weighted avg      0.836     0.833     0.832       257
```

Figure 1. Model Evaluation Metrics – displays the model's performance using Accuracy, Precision, Recall, and F1-score, providing a summary of predictive results

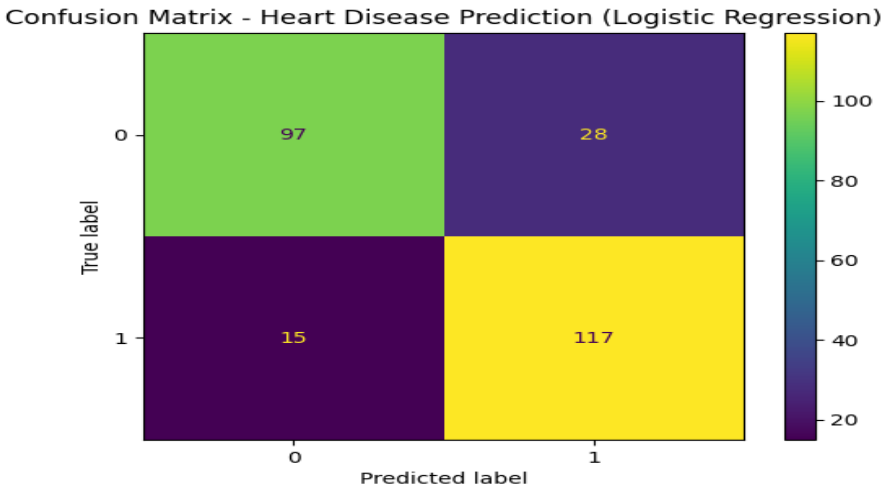


Figure 2. Confusion Matrix Heatmap – shows the number of correct and incorrect predictions for each class.

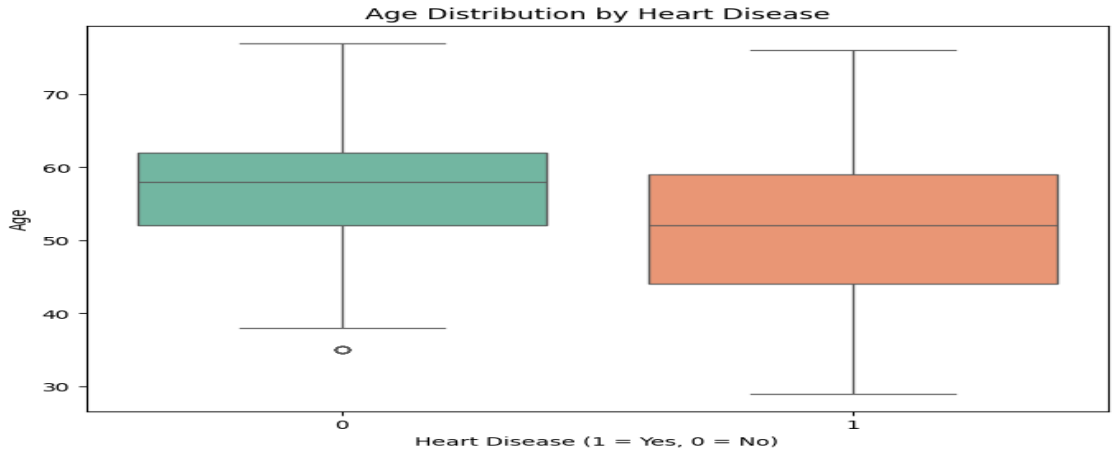


Figure 3. Age Distribution by Heart Disease – displays the age spread across patients with and without heart disease.

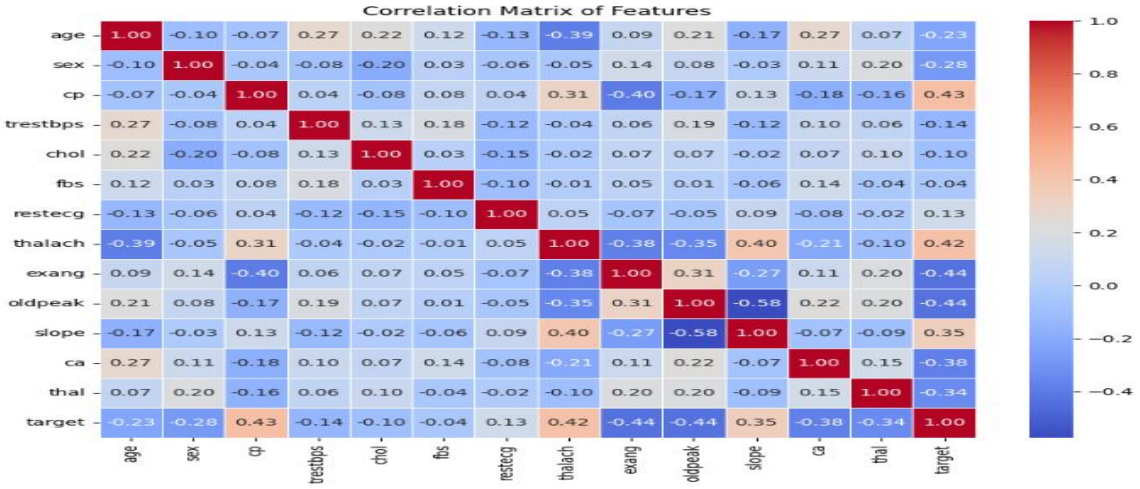


Figure 4. Correlation Matrix of Features – visualizes relationships among medical features and the target.

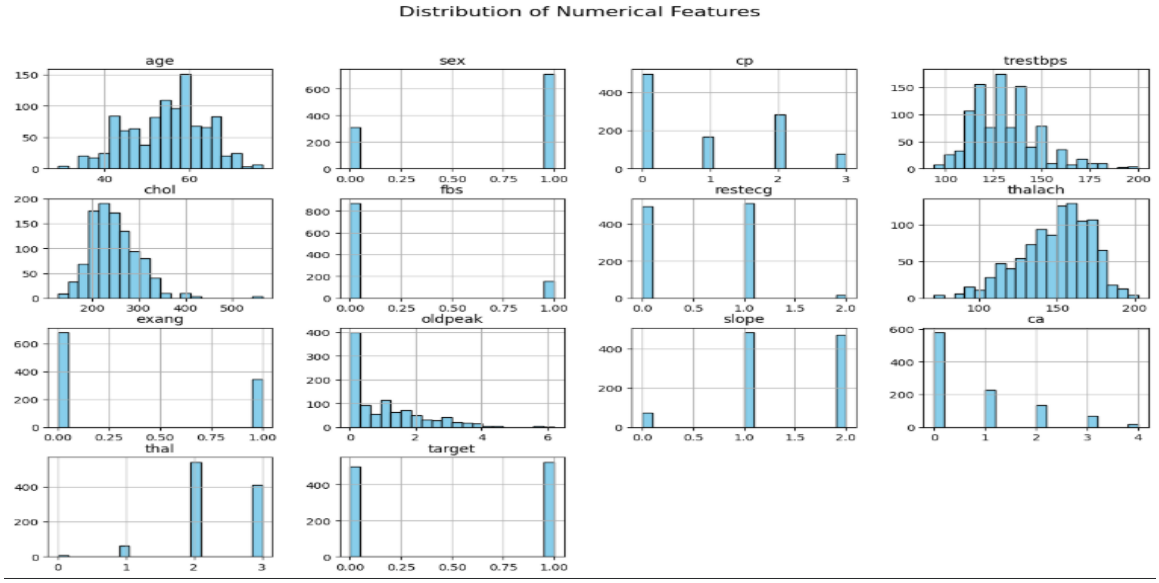


Figure 5. Distribution of Numerical Features – shows histogram plots of numeric variables.